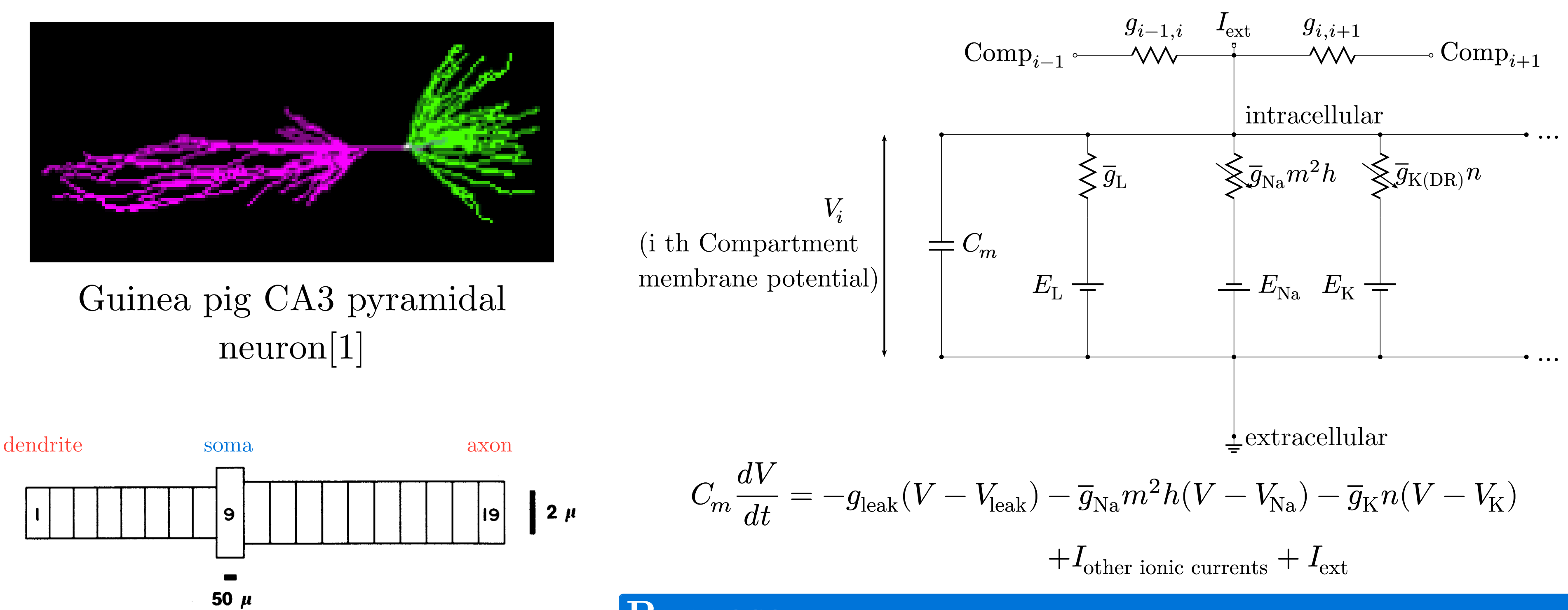


Introduction



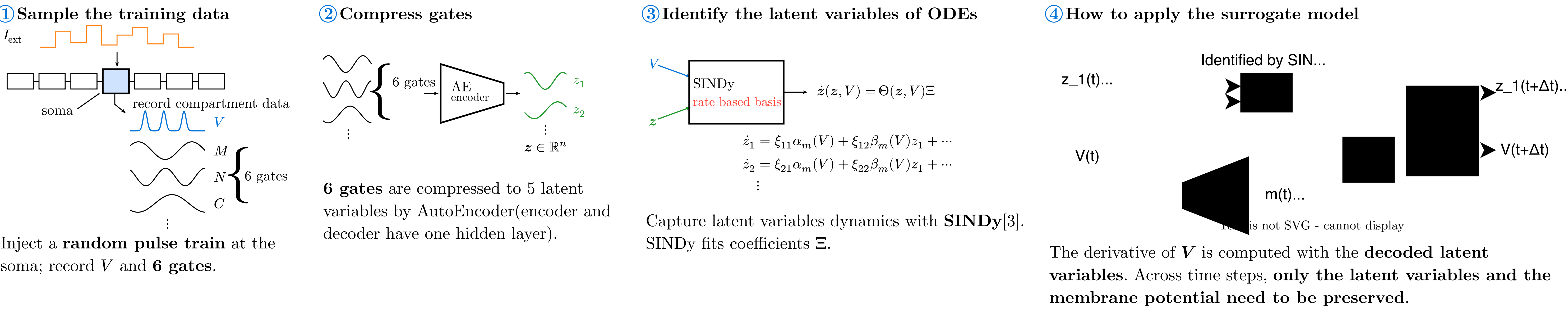
The conductances have **gate variables**, ODEs with rate functions α, β :

$$I_{Na} = \bar{g}_{Na} m^2 h (V - E_{Na})$$
$$\frac{dm}{dt} = \alpha_m(V)(1 - m) - \beta_m(V)m$$
$$\alpha_m(V) = 0.32(13.1 - u)/(\exp((13.1 - u)/4) - 1)$$
$$\beta_m(V) = 0.28(u - 40.1)/(\exp((u - 40.1)/5) - 1)$$

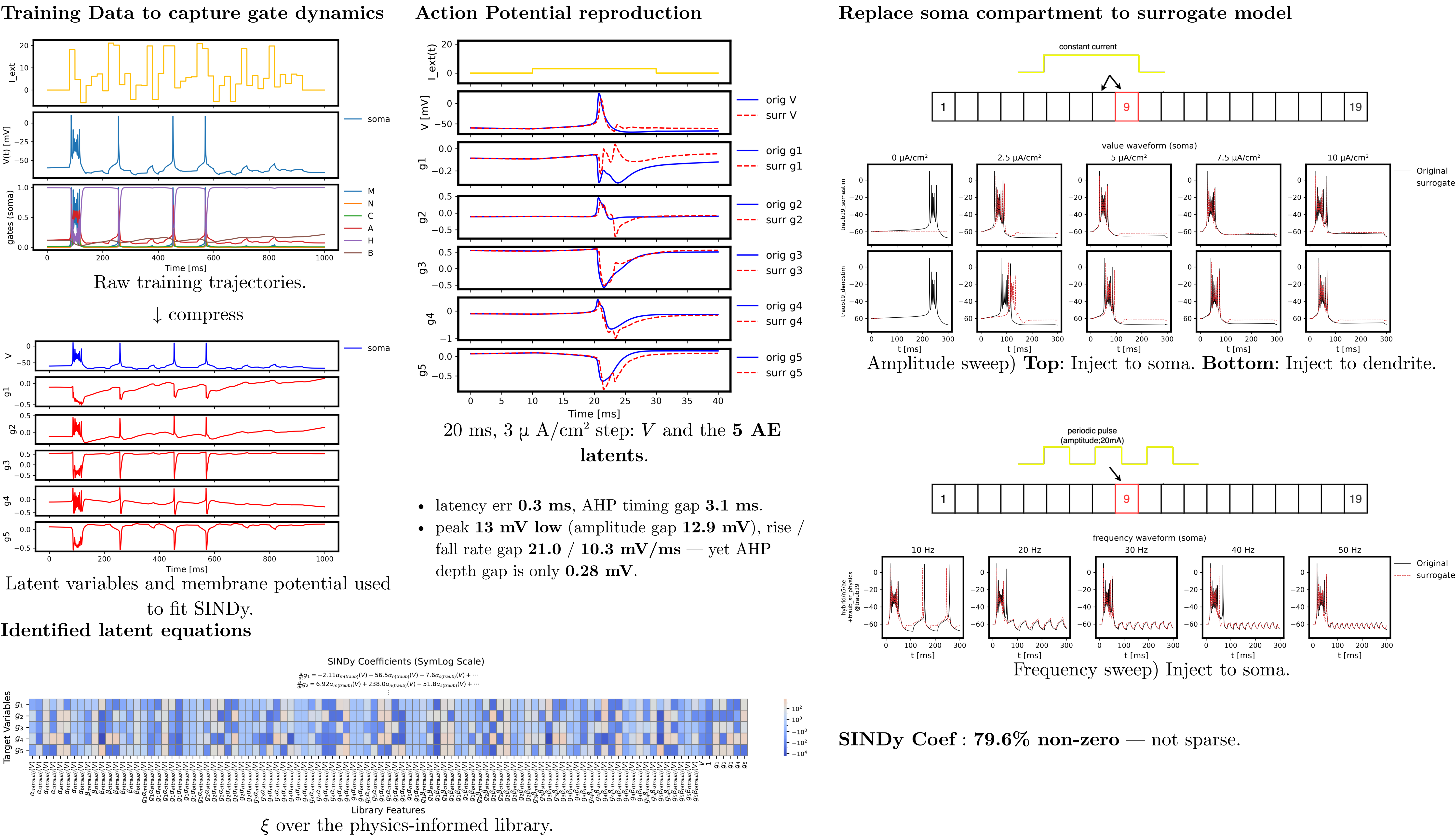
→ **11 states (10 gates and V) per comp**
→ **209 states** for 19 comps;
In large scale network simulation, number of large gate variables become a **memory bottleneck**.

Purpose
Development of a multi-compartment neuron’s surrogate model capable of reproducing the membrane potential response with fewer gate variables.

Methods



Results and Discussion



→ **Future:** sparsify ξ , push n down, test on other channel models.

Conflicts of Interest — The authors declare no conflicts of interest regarding this manuscript.